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18/417,481	01/19/2024	Yu Fang	SONA-0015-U01-C05	6428

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EXAMINER
ALGIBHAH, HAMZA N

ART UNIT	PAPER NUMBER
2441	

NOTIFICATION DATE	DELIVERY MODE
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Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

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Office Action Summary	Application No. 18/417,481	Applicant(s) Fang et al.	
	Examiner HAMZA N ALGIBHAH	Art Unit 2441	AIA (FITF) Status Yes

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address --

Period for Reply

A SHORTENED STATUTORY PERIOD FOR REPLY IS SET TO EXPIRE 3 MONTHS FROM THE MAILING DATE OF THIS COMMUNICATION.

- Extensions of time may be available under the provisions of 37 CFR 1.136(a). In no event, however, may a reply be timely filed after SIX (6) MONTHS from the mailing date of this communication.
- If NO period for reply is specified above, the maximum statutory period will apply and will expire SIX (6) MONTHS from the mailing date of this communication.
- Failure to reply within the set or extended period for reply will, by statute, cause the application to become ABANDONED (35 U.S.C. § 133). Any reply received by the Office later than three months after the mailing date of this communication, even if timely filed, may reduce any earned patent term adjustment. See 37 CFR 1.704(b).

Status

1)☑ Responsive to communication(s) filed on 01/09/2024.

☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on _____.

2a)☐ This action is **FINAL.** 2b) ☑ This action is non-final.

3)☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on _____; the restriction requirement and election have been incorporated into this action.

4)☐ Since this application is in condition for allowance except for formal matters, prosecution as to the merits is closed in accordance with the practice under *Ex parte Quayle*, 1935 C.D. 11, 453 O.G. 213.

Disposition of Claims*

5) ☑ Claim(s) 1-16 is/are pending in the application.

5a) Of the above claim(s) _____ is/are withdrawn from consideration.

6) ☐ Claim(s) _____ is/are allowed.

7) ☑ Claim(s) 1-6 and 9-14 is/are rejected.

8) ☑ Claim(s) 7-8 and 15-16 is/are objected to.

9) ☐ Claim(s) _____ are subject to restriction and/or election requirement

* If any claims have been determined allowable, you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to **PPHfeedback@uspto.gov.**

Application Papers

10)☐ The specification is objected to by the Examiner.

11)☑ The drawing(s) filed on 01/09/2024 is/are: a)☑ accepted or b)☐ objected to by the Examiner.

Applicant may not request that any objection to the drawing(s) be held in abeyance. See 37 CFR 1.85(a).

Replacement drawing sheet(s) including the correction is required if the drawing(s) is objected to. See 37 CFR 1.121(d).

Priority under 35 U.S.C. § 119

12)☐ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

a)☐ All b)☐ Some** c)☐ None of the:

1.☐ Certified copies of the priority documents have been received.

2.☐ Certified copies of the priority documents have been received in Application No. _____.

3.☐ Copies of the certified copies of the priority documents have been received in this National Stage application from the International Bureau (PCT Rule 17.2(a)).

** See the attached detailed Office action for a list of the certified copies not received.

Attachment(s)

1) ☑ Notice of References Cited (PTO-892)

2) ☑ Information Disclosure Statement(s) (PTO/SB/08a and/or PTO/SB/08b) Paper No(s)/Mail Date _____.

3) ☐ Interview Summary (PTO-413) Paper No(s)/Mail Date _____.

4) ☐ Other: _____.

Detailed Action

1. ***Claims 1-16*** are pending.

2. ***Claims 1-6 and 9-14 are rejected.***

3. ***Claims 7-8 and 15-16 are objected to.***

Claim Rejections - 35 USC § 103

The following is a quotation of 35 U.S.C. 103 which forms the basis for all obviousness rejections set forth in this Office action:

> A patent for a claimed invention may not be obtained, notwithstanding that the claimed invention is not identically disclosed as set forth in section 102, if the differences between the claimed invention and the prior art are such that the claimed invention as a whole would have been obvious before the effective filing date of the claimed invention to a person having ordinary skill in the art to which the claimed invention pertains. Patentability shall not be negated by the manner in which the invention was made.

4. ***Claims 1-6 and 9-14 are rejected*** under 35 U.S.C. 103 as being unpatentable over MOHAMMADI et al (Pub. No.: US 2022/0217074 A1) in view of BARRETT et al (Pub. No.: US 2019/0394089 A1) and Wang et al (Pub. No.: US 2022/0158974 A1).

As per claim 1, MOHAMMADI discloses a system, comprising:

- ***a ~~vehicle~~ multi-zone monitoring platform*** (MOHAMMADI, paragraph 0057, wherein each network node's topology graph and link state database can consist of multiple portions, which may be referred to as zones, levels, or regions)***, comprising:***

a network layout circuit structured to interpret a virtual network layout ~~for a vehicle~~ having a multi-zone network (MOHAMMADI, Fig 3, 5, paragraph 0081, 0083, wherein the multi-zone/multi-precision network status information component 540 is configured to maintain network status information for a plurality of different portions (zones) of the network. Network

status information for different portions of the network has different levels of precision)***;***

- ***a network characterization circuit structured to interpret network activity ~~for the vehicle~~*** (MOHAMMADI, Fig 3, 5, paragraph 0082, wherein the network interface can receive flooding status update messages 560a indicative of network events, almanac status update messages 560b indicative of (e.g. hourly based) network topology information constructed based on network events, network status updates 560c from terrestrial ground stations which monitor network conditions and transmit such update messages, or a combination thereof. The different updates 560a, 560b, 560c may be received in different time intervals and may cover different numbers of nodes or portions of the network)***; and***

- ***a network analysis circuit structured to determine a virtual network map in response to the virtual network layout and the network activity*** (MOHAMMADI, Fig 3, 5, paragraph 0084, wherein the routing information generation component 545 is configured to generate routing information based on said network status information maintained by the multi-zone/multi-precision network status information component 540, for the plurality of different portions of the network in combination. This can include generating shortest path trees specifying packet routing policies, building routing tables, etc. The routing information generation component can merge shortest path trees for multiple zones together. The routing information generation component can merge network topology graphs for multiple zones, and generate shortest path trees for multiple zones based on the merged network topology graph). MOHAMMADI does not explicitly disclose that the multi-zone monitoring platform is a ***vehicle*** multi-zone monitoring platform and ***to provide the virtual network map to a user interface***. However, BARRETT discloses that the multi-zone monitoring platform is a ***vehicle*** multi-zone monitoring platform (BARRETT, Fig 1-2, paragraph 0021, wherein The vehicle network 100 can include various network nodes that enable secure and reliable communications among the vehicle's electronic

systems. The network node can include application software that can be executed by one or more processors. The network nodes can include ECUs (e.g., ECUs 103, 113, 123, 143, 153), domain controllers (DCs, e.g., DCs 133), sensors, actuators, or any other types of electronic devices or computers that are capable of data communications. Different network nodes can have different network interfaces that communicate over different network protocols. In some implementations, the internal network 100 can be a heterogeneous network. For example, a vehicle can integrate over 100 ECUs connected over multiple networks such as CAN (Control Area Network), LIN (Local Interconnect Network), FlexRay, and Ethernet).

Therefore, it would have been obvious to one ordinary skill in the art before the effective filing date of the invention to modify MOHAMMADI in view of BARRETT to support packet forwarding and routing in a vehicle environment because this would have provided a way to maintain awareness of the current and future available network connections, in the form of network status information in a vehicle environment.

MOHAMMADI and BARRETT do not explicitly disclose ***to provide the virtual network map to a user interface***. However, Wang discloses ***to provide the virtual network map to a user interface*** (Wang, Fig 6E, paragraph 0136-0137, wherein the Internet-of-things social network graph display screen 646 presents various information associated with the example Internet-of-things network 104. In particular, the Internet-of-things social network graph display screen 646 shown in FIG. 6E shows multiple node icons 650, which can represent nodes on the Internet-of-things network 104).

Therefore, it would have been obvious to one ordinary skill in the art before the effective filing date of the invention to modify MOHAMMADI and BARRETT in view of Wang so that a network map is displayed in a user interface as claimed because this would have provided

a way to maintain awareness of the current and future available network connections, in the form of network status information in a vehicle environment.

As pre claim 2, claim 1 is incorporated and Wang discloses wherein the virtual network map comprises an interactive network activity depiction (Wang, Fig 6A-6E, paragraph 0138-0139, wherein the Internet-of-things social network graph display screen 646 also depicts connections or links ("links") 652 between the Internet-of-things devices 102 represented by the node icons 650. These links 652 can illustrate how the Internet-of-things devices 602 represented by the node icons 650 communicate with one another. The node icons 650, links 652, and alert links 654 in the Internet-of-things social network graph display screen 646 can correspond to UI features, so tapping or clicking these or other elements may cause a display of additional information associated with the respective links and/or Internet-of-things devices 102).

As pre claim 3, claim 2 is incorporated and Wang discloses wherein the network analysis circuit is further structured to provide a network detail value in response to a user selection on the user interface (Wang, Fig 6A-6E, paragraph 0138-0139, wherein the Internet-of-things social network graph display screen 646 also depicts connections or links ("links") 652 between the Internet-of-things devices 102 represented by the node icons 650. These links 652 can illustrate how the Internet-of-things devices 602 represented by the node icons 650 communicate with one another. The node icons 650, links 652, and alert links 654 in the Internet-of-things social network graph display screen 646 can correspond to UI features, so tapping or clicking these or other elements may cause a display of additional information associated with the respective links and/or Internet-of-things devices 102).

As pre claim 4, claim 3 is incorporated and Wang discloses **wherein the network detail value comprises at least one of: a network traffic description; a zoned architecture communication scheme; a network statistics value; or a network event notification** (Wang, Fig 6A-6E, paragraph 0138-0139, wherein the Internet-of-things social network graph display screen 646 also depicts connections or links ("links") 652 between the Internet-of-things devices 102 represented by the node icons 650. These links 652 can illustrate how the Internet-of-things devices 602 represented by the node icons 650 communicate with one another. The node icons 650, links 652, and alert links 654 in the Internet-of-things social network graph display screen 646 can correspond to UI features, so tapping or clicking these or other elements may cause a display of additional information associated with the respective links and/or Internet-of-things devices 102).

As pre claim 5, claim 1 is incorporated and MOHAMMADI discloses **wherein the vehicle multi-zone monitoring platform further comprises a network configuration circuit structured to interpret a network adjustment value, and to update the virtual network map in response to the network adjustment value** (MOHAMMADI, paragraph 0086, wherein network status information maintenance operations for additional network portions or precision levels may also be performed. In some embodiments, network status information for different network portions can be maintained by processing different types of network status update messages, with different latencies or levels of precision. Maintaining network status information for a particular portion of the network can include maintaining a network topology graph for that network portion, which is adjusted based on network status update messages).

As pre claim 6, claim 5 is incorporated and MOHAMMADI discloses wherein the network analysis circuit is further structured to determine a network impact value in response to the network adjustment value, and to provide the network impact value to the user interface (MOHAMMADI, paragraph 0086, wherein network status information maintenance operations for additional network portions or precision levels may also be performed. In some embodiments, network status information for different network portions can be maintained by processing different types of network status update messages, with different latencies or levels of precision. Maintaining network status information for a particular portion of the network can include maintaining a network topology graph for that network portion, which is adjusted based on network status update messages)

Claims 9-14 are rejected under the same rationale as claim 1-6.

Allowable Subject Matter

Claims 7-8 and 15-16 are objected to as being dependent upon a rejected base claim, but would be allowable if rewritten in independent form including all of the limitations of the base claim and any intervening claims.

Conclusion

Any inquiry concerning this communication or earlier communications from the examiner should be directed to HAMZA N ALGIBHAH whose telephone number is (571)270-7212. The examiner can normally be reached 7:30 am - 3:30 pm.

Examiner interviews are available via telephone, in-person, and video conferencing using a USPTO supplied web-based collaboration tool. To schedule an interview, applicant is encouraged to use the USPTO Automated Interview Request (AIR) at http://www.uspto.gov/interviewpractice.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, Wing Chan can be reached on (571) 272-7493. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

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/HAMZA N ALGIBHAH/
Primary Examiner, Art Unit 2441